Friction Investigation Planning Sheet

1. PLANNING

1.1 Aim

What are you trying to find out? What problem are you trying to solve?

_____

_____

_____

1.2 Variables

You can only change one variable at a time and measure its effect. Select one that you would like to investigate. The rest of the variables must be kept the same.

Change (Independent)	**Measure (Dependent)**	**Keep the Same (Controlled)**
_____	_____	_____
		_____
		_____

1.3 Hypothesis

A statement about what you think will happen. This statement needs to link the independent and dependent variables.

_____

_____

_____

2. CONDUCTING AND RECORDING

2.1 Equipment

Write a list of all the equipment you will need to conduct your experiment.

- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____
- _____

2.2 Procedure

Design a method by which you can slowly change one variable and measure its affect on the other.

1. _____
   _____
2. _____
   _____
3. _____
   _____
4. _____
   _____
5. _____
   _____
6. _____
   _____
7. _____
   _____
8. _____
   _____

2.3 Results

Construct a table so that you can record your results.

3. ANALYSING

3.1 Analysing Your Results

Do your results support your hypothesis or not?

_____

_____

Explain **how** you used your results to come to your conclusion above.

_____

_____

_____

_____

_____

4. EVALUATION

4.1 Errors

What are the possible sources of error in your investigation?

_____

_____

_____

How might these sources of error affect your data?

_____

_____

_____

4.2 Improving Your Investigation

What changes would you make to your investigation to improve your results?

_____

_____

_____

_____

_____

_____

5. REPORT

You need to write a report of your experiment. Your report should include the following:

Aim: What is the aim or purpose of your investigation?

Hypothesis: Your prediction in the form of a statement. This needs to link your independent and dependent variables.

Equipment: A list of all the equipment you used in your investigation.

Diagram of Equipment: A labelled diagram of how you set up your equipment to conduct your investigation.

Method: A list of the steps you went through to conduct your investigation.

Results: The results of your investigation presented in a table.

Conclusion: Comment on

- Whether or not your results supported your hypothesis and how you came to that conclusion
- What were the possible sources of error in the investigation
- What you could do to improve your investigation.

Your report will be marked according to the following:

- Aim (2 marks)
- Hypothesis (2 marks)
- Equipment (2 marks)
- Diagram of Equipment (2 marks)
- Method (2 marks)
- Results (2 marks)
- Conclusion (6 marks)
- Presentation (1 for neatness, 1 for appropriate use of headings)
- Total (20 marks)